**Solar ESS Co. Ltd.**
53 Moo 2 Tambon Nanglae, Amphoe Muang, Chiang Rai 57100
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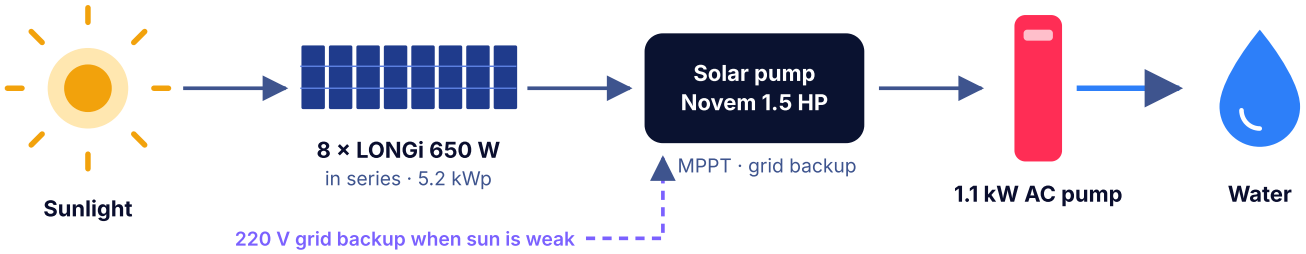
SOLAR WATER PUMPING · SYSTEM 1 · BUDGET BOM & PRICE

# Hilltop tank system

Borehole 30–40 m, water pumped 30 m uphill over 200 m into a tank on the hill. A TORQUE 1.1 kW borehole pump, sized for this head, is driven by a Novem ND1-2-003-S solar pump inverter from 8 × LONGi 650 W panels, with 220 V grid power as backup when the sun is weak.

TN-20261019-PUMP-03-S1 | Rev07 | 20 October 2026 | Chiang Rai, Thailand | Prices in THB, confirmed at order

## How it works




### 1 Hilltop tank

Borehole 30–40 m, 200 m uphill pipe to a tank 30 m above the well.

| ≈ 70 m | 12–18 m³ | 200 m | 2–3 m³/h |
|---|---|---|---|
| Total dynamic head | Water per sunny day* | Uphill pipe run | Expected flow* |

SYSTEM 1 PRICE
**120,850** THB excl. VAT
129,310 THB including 7% VAT

## The pump · TORQUE TQ-SP-2DLBM20-4/S

| | | | |
|---|---|---|---|
| **1.1 kW**<br>1.5 HP motor | **220 V**<br>1-phase AC | **130 m**<br>Max head | **53–119 m**<br>Working head range |
| **1½"**<br>Outlet | **20**<br>Impellers · 4" well | **8 × 650 W**<br>LONGi panels · 5.2 kWp | **Novem ND1-2-003-S**<br>1.5 HP · 11 A pump inverter |

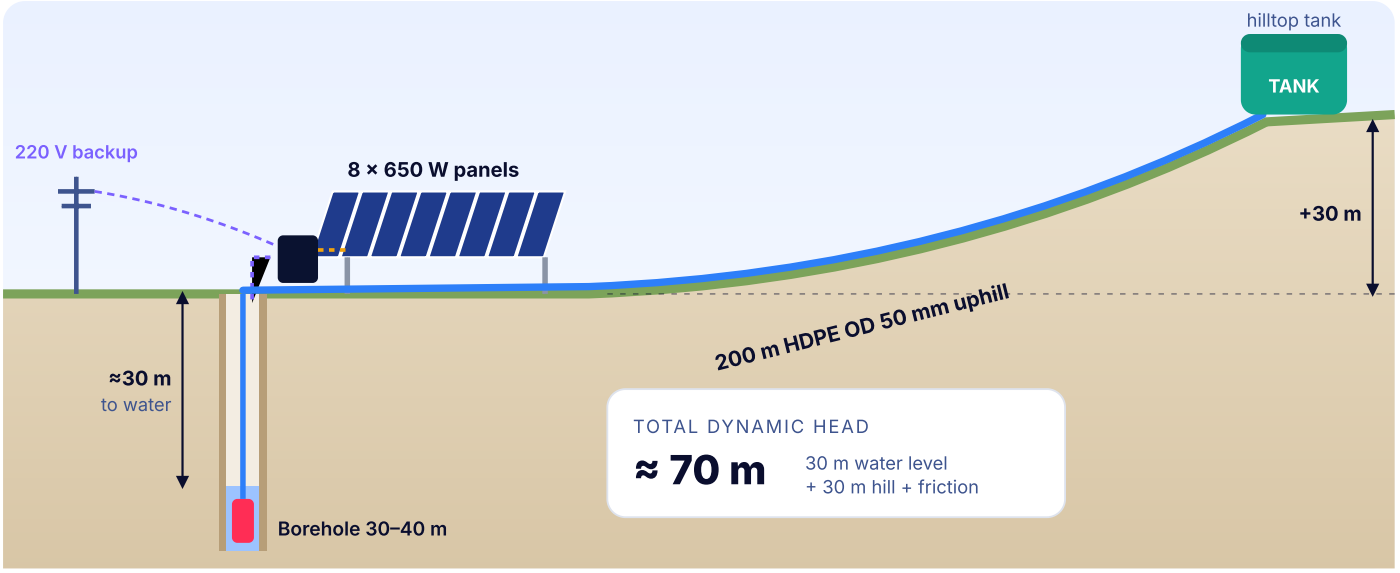
* Flow and daily volume are estimates: the maker does not publish flow figures for this model. They will be confirmed with the pump curve at 70 m before the final quotation.

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SYSTEM 1 · THE SITE

# Borehole 30–40 m, pumped 30 m uphill over 200 m

Total dynamic head ≈ 70 m (water level ~30 m + 30 m hill + friction). Expected output about 2–3 m³/h into the hilltop tank, roughly 12–18 m³ per sunny day, to be confirmed with the pump curve.




| HEAD | FLOW | PER SUNNY DAY | PUMP CURVE |
|---|---|---|---|
| ≈ 70 m | 2–3 m³/h | 12–18 m³ | To confirm |
| water level + hill + friction | estimate, at ≈ 65–70 m head | 12,000–18,000 L | flow at 70 m before final quote |

## Why this pump: which TORQUE pump works best at 70 m?

| PUMP | WORKING HEAD RANGE (M) · DASHED LINE = THIS SITE, 70 M | FIT |
|---|---|---|
| TQ-SP-2DLBM14-4/S<br>1.0 HP · 15,000 THB | 37–83 m | Near top of range |
| **TQ-SP-2DLBM20-4/S**<br>1.5 HP · 17,000 THB | 53–119 m | **Best fit · selected** |
| TQ-SP-2DLBH28-4/S<br>2.0 HP · 21,000 THB | 74–167 m | Below its range |
| TQ-SP-2DLBH40-4/S<br>3.0 HP · 29,000 THB | 106–238 m | Far below · oversized |

Each bar is the head range the maker recommends for that pump; the thin line is its maximum. The 1.5 HP TQ-SP-2DLBM20-4/S is the only one with 70 m comfortably inside its range: enough reserve for the hill and the 200 m pipe, without the larger pumps running off the end of their curves and wasting power.

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SYSTEM 1 · BILL OF MATERIALS

# Hilltop tank system · price

| TOTAL EXCL. VAT | VAT 7% | TOTAL INCL. VAT | OPTIONAL TANK |
|---|---|---|---|
| **120,850** THB | **8,460** THB | **129,310** THB | **≈ 9,500** 3,000 L PE + base, not in total |

## Where the money goes




Pump & inverter **27,500** · Solar array & frame **46,400** · Piping & valves **15,300** · Electrical & protection **10,850**
Safety accessories **800** · Installation **20,000**

## Itemised list

| # | ITEM | QTY | UNIT (THB) | AMOUNT (THB) |
|---|---|---|---|---|
| 1 | TORQUE TQ-SP-2DLBM20-4/S borehole pump, 1.1 kW (1.5 HP), 1-phase 220 V, 1½" outlet, 20 impellers, for 4" well | 1 set | 17,000 | **17,000** |
| 2 | Novem ND1-2-003-S solar pump inverter, 1.5 HP 1-phase 220 V output, 11 A, MPPT 250–450 VDC, hybrid PV + 220 V grid input, dry-run protection (budget price) | 1 | 10,500 | **10,500** |
| 3 | LONGi 650 W solar panel | 8 | 3,800 | **30,400** |
| 4 | Galvanised ground-mount frame, 8 panels | 1 set | 16,000 | **16,000** |
| 5 | HDPE pipe OD 50 mm PN10 (well drop + 200 m uphill line) | 240 m | 45 | **10,800** |
| 6 | HDPE fittings, 1½"→OD50 adaptor, check valve, gate valve | 1 lot | 4,500 | **4,500** |
| 7 | Submersible pump cable 4 × 2.5 mm² | 50 m | 55 | **2,750** |
| 8 | PV cable 4 mm² red/black + MC4 connectors | 1 lot | 2,000 | **2,000** |
| 9 | DC breaker + DC SPD + steel enclosure | 1 set | 3,500 | **3,500** |
| 10 | AC breaker for grid backup supply | 1 | 800 | **800** |
| 11 | Earth rod and earth cable | 1 set | 1,200 | **1,200** |
| 12 | Tank float switch | 1 | 600 | **600** |
| 13 | Stainless safety rope + pump clamps + well cap | 1 set | 800 | **800** |
| 14 | Installation, pipe laying and commissioning | 1 lot | 20,000 | **20,000** |
| | **Total (excluding VAT)** | | | **120,850** |
| | VAT 7% | | | 8,460 |
| | **Total including VAT** | | | **129,310** |

## Notes

฿ Prices are budget figures excluding VAT unless stated. Pump and inverter prices are from current market listings; other items are market estimates. Final prices are confirmed at order.

↗ The maker publishes maximum head (130 m) and working range (53–119 m) but no flow figures. At about 70 m the pump sits near the middle of its range, where a borehole pump is most efficient. The 2–3 m³/h estimate comes from the motor power and head; the pump curve will be obtained to confirm it before the final quotation.

≈ The Novem ND1-2-003-S (11 A) is rated for 1.5 HP single-phase 220 V motors, matching the 1.1 kW pump (about 7–8 A). 8 × 650 W in series (≈ 300 V MPP, ≈ 395 V Voc) stays inside its 250–450 VDC range even on hot days, and the extra array power starts the pump earlier and runs it longer. Its price and how it switches to grid power are confirmed with Novem before order.

⚡ All electrical work uses copper cable, proper DC protection and earthing, installed by a certified electrician.